

Daniel (Chenyu) Lin

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EDUCATION

Cornell University

Ithaca, NY

B.A. in Computer Science and Mathematics (double major)

Aug 2024 – Expected May 2027

University of North Carolina at Chapel Hill

Chapel Hill, NC

Transferred Out, Computer Science

Aug 2023 – May 2024

Selected Coursework:

AI/ML: Machine Learning | Large-Scale ML | Natural Language Processing | Reinforcement Learning (ongoing)
Maths: Information Theory (grad) | Measure Theory (grad, ongoing) | Stochastic Processes | (Advanced) Linear Algebra | Honors Analysis

RESEARCH INTERESTS

My research centers on **unsupervised and self-supervised learning**, with **music and audio technologies** as my primary downstream domain. I am drawn to systems that discover structure on their own, and to what such systems can offer creative work that more heavily supervised approaches cannot. I am also interested in the theory underlying these methods, and in the mathematical structure that determines what unsupervised learning can recover.

PUBLICATIONS

- Daniel Chenyu Lin, Michael Freeman, John Thickstun. “Assessing Factual Music Comprehension in Large Audio Language Models.” *International Society for Music Information Retrieval (ISMIR)*, 2026.
- Louis Bradshaw, Irmak Bukey, Daniel Chenyu Lin, John Thickstun, Chris Donahue. “PianoBench: Evaluating Musical Perception and Reasoning in Language Models.” *In submission to ICLR 2027*.
- Saebyeol Shin, Chao Wan, Zhenzhen Liu, Justin Lovelace, Daniel C. Lin, Kilian Q. Weinberger, John Thickstun. “Music Transcription with (Almost) No Supervision.” *Preprint*.

RESEARCH EXPERIENCE

Cornell Bowers CIS | Advisor: Dr. John Thickstun

Ithaca, NY

Student Researcher, Generative Models & Music Information Retrieval

Oct 2024 – Present

Oct 2024 – Apr 2026

- **Factual evaluation of large audio language models**
Showed that standard text-similarity evaluation on MusicQA cannot distinguish LALM responses conditioned on correct audio from responses conditioned on random audio. Designed a factual evaluation protocol that parses open-ended responses into structured labels scored by precision/recall/F1, and benchmarked nine frontier and open-source models across six factual retrieval tasks on MusicNet, FMA, and a newly curated dataset. Awarded a \$6,000 research grant from Cornell’s College of Arts and Sciences. (First author; accepted to ISMIR 2026)
- **Unsupervised automatic music transcription** | Co-advisor: Dr. Kilian Q. Weinberger Oct 2025 – May 2026
Co-authored a study of unsupervised transcription with light supervision to break the symmetry on symbolic marginals; built the training corpus, scraping and normalizing ten sources into ~440 hours of audio and ~57 hours of symbolic data across four annotation formats. (Preprint)
- **Symmetry and identifiability in unsupervised translation** May 2026 – Aug 2026
Led the theoretical analysis of why unsupervised transcription recovers the target map up to a constant pitch shift, characterizing the symmetry group of the symbolic marginal and deriving deterministic and variational cycle-consistent objectives with gradient penalties for a controlled 2D translation testbed, then analyzing why these objectives fail to isolate the intended solution.
- **Instruction tuning LLMs for MIDI understanding** | Co-advisor: Dr. Chris Donahue Aug 2026 – Present
Built an instruction-tuning pipeline supporting full fine-tuning, LoRA, and DoRA for LLMs following symbolic music mid-training. Evaluated music understanding, general knowledge, and instruction following using PianoBench, MMLU-Redux, and IFEval. Combined music mid-training and instruction tuning increased PianoBench overall multiple-choice accuracy from 40.4% to 55.2% relative to the base checkpoint. Instruction tuning recovered losses in general knowledge and instruction following while largely preserving musical understanding. (In submission to ICLR 2027)

Lyra Lab, Tencent Music Entertainment

Shenzhen, China

Research Intern, Audio Information Retrieval

Jun 2024 – Aug 2024

- Replaced the Conformer encoder in a production Cover Song Identification system with a Zipformer encoder, raising top-1 accuracy by 7 points to over 90%, and resolved exploding gradients at scale by reimplementing the block's activation function.
- Pre-trained MT3 from scratch on roughly twice the data used in the original paper, improving instrument detection and onset alignment.
- Built an in-batch audio augmentation pipeline supporting speed and pitch variation, noise injection, and impulse response convolution.

Music Tech X (High School Program) Student Researcher & Group Leader

Jun 2023 – Aug 2023

- Implemented an audio fingerprinting algorithm from scratch for music recognition, and led a team of six in improving its search and evaluation components to reach 96% precision on the MIREX dataset.
- Explored a learned alternative to fingerprinting, training a ResNet to recognize pitch-shifted and tempo-varied music with over 80% precision.

TEACHING & ACADEMIC INVOLVEMENT

Reading Group Participant & Presenter

Cornell University & Ithaca College

Ithaca, NY

- Theory of Generative Modeling (CS 7792)
- Music Technology

Spring 2026

Fall 2024 – Spring 2025

Teaching Assistant

Cornell Bowers CIS

Ithaca, NY

- ECE 5620/ECE 7620: Information Theory and Generative Modeling
- CS 4782/CS 5782: Introduction to Deep Learning
- CS 3780/CS 5780: Introduction to Machine Learning
- CS 1109: Foundational Programming Concepts

Fall 2026

Spring 2026

Fall 2025

Summer 2025

TECHNICAL SKILLS

Programming Languages: Python, Java, OCaml, C (cuBLAS), ARM Assembly

Machine Learning: Pytorch, Tensorflow, Huggingface, W&B, Data Augmentations

Audio Programming: Librosa, Torchaudio, Madmom, Muscore, Logic, Audacity

Other Useful Skills: Linux, Slurm Cluster, GPU Deployment, Github, Google Colab